

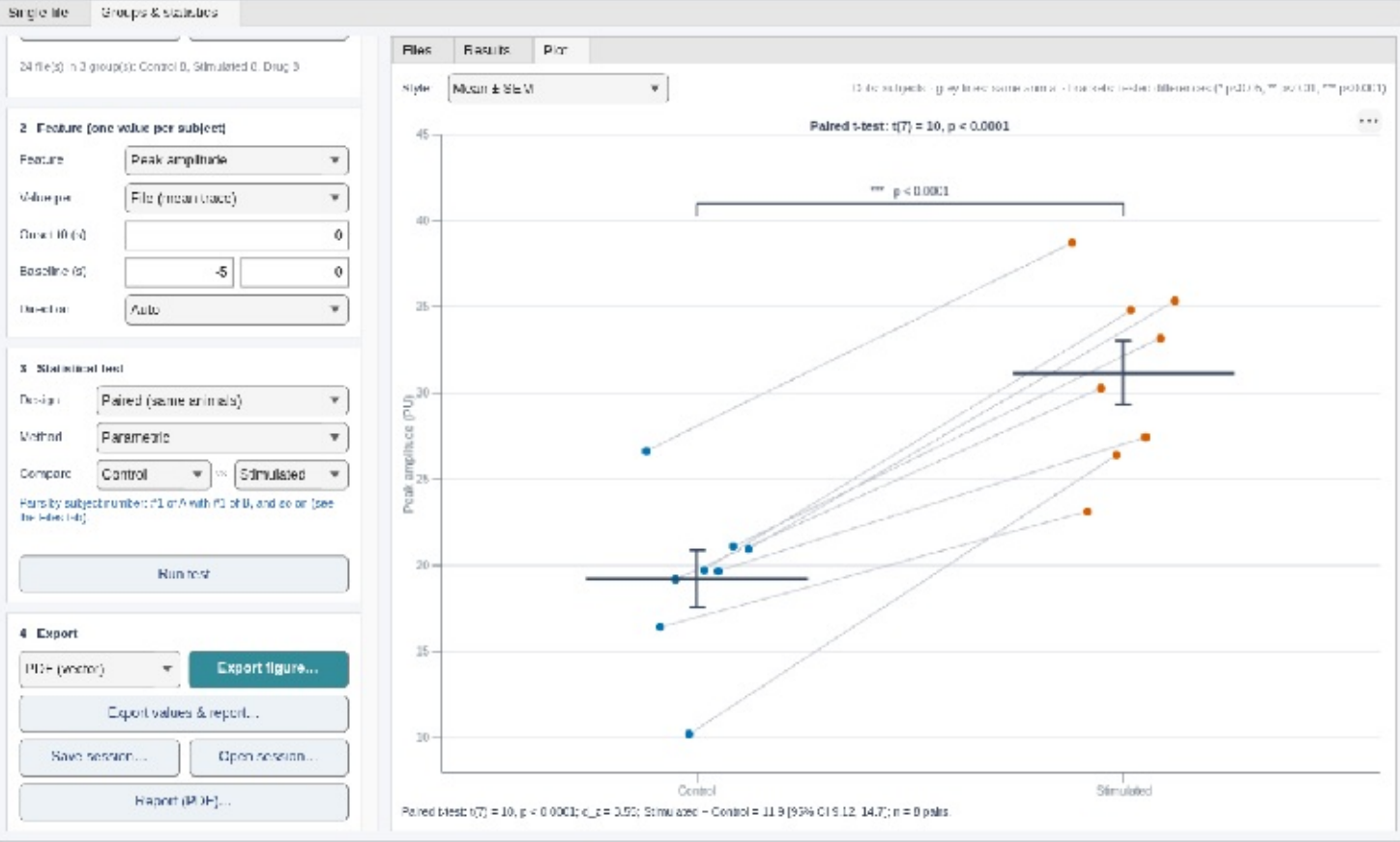
NeuroAnalyzer report: Signal Characterization

NeuroAnalyzer v0.3.0 · MATLAB 2026a · 2026-09-25T03:22:00+00:00 · window image: exportapp

Signal Characterization

Default required features (feature, FWHM, AUC, rise/decay) will compare groups with statistics

Help



Window: Signal Characterization (SignalCharacterizationApp)
NeuroAnalyzer v0.3.0
MATLAB 26.1.0.3346908 (R2026a) Update 5
OS: Linux 6.17.0-1022-azure (GLNXA64)
Session created: 2026-09-25T03:22:00+00:00
Report created: 2026-09-25T03:22:00+00:00

NOTES
SignalCharacterizationApp walkthrough: demo data

INPUTS (25)
1. Single file: demo_ldf_trials.mat
path: /tmp/NeuroAnalyzerDemo/demo_ldf_trials.mat
30,187 bytes, modified 2026-09-25T03:16:48
MD5 ba186b0fc9c12a54330fcb9d44a15a92
2. Group Control #1: control_animal01.mat
path: /tmp/NeuroAnalyzerDemo/groups/control_animal01.mat
15,806 bytes, modified 2026-09-25T03:21:54
MD5 eb898aca7d089830186f6cddc129442e
3. Group Control #2: control_animal02.mat
path: /tmp/NeuroAnalyzerDemo/groups/control_animal02.mat
15,783 bytes, modified 2026-09-25T03:21:54
MD5 6cbdea505ff3d5f9911f3294123b9863
4. Group Control #3: control_animal03.mat
path: /tmp/NeuroAnalyzerDemo/groups/control_animal03.mat
15,764 bytes, modified 2026-09-25T03:21:54
MD5 37a575d23fa5c330941523335e99210f
5. Group Control #4: control_animal04.mat
path: /tmp/NeuroAnalyzerDemo/groups/control_animal04.mat
15,730 bytes, modified 2026-09-25T03:21:54
MD5 af80993a6b34ca0e9a9e3711952b4d24
6. Group Control #5: control_animal05.mat
path: /tmp/NeuroAnalyzerDemo/groups/control_animal05.mat
15,769 bytes, modified 2026-09-25T03:21:54
MD5 342a01aa416880f0a784a9c30f21b561
7. Group Control #6: control_animal06.mat
path: /tmp/NeuroAnalyzerDemo/groups/control_animal06.mat
15,669 bytes, modified 2026-09-25T03:21:54
MD5 7322c060a103fa35c0ebf6ccfd5c5de10
8. Group Control #7: control_animal07.mat
path: /tmp/NeuroAnalyzerDemo/groups/control_animal07.mat
15,677 bytes, modified 2026-09-25T03:21:54
MD5 e3364c89f187eb0ab3891143a394226a
... 68 more line(s): see the session file

SETTINGS
mode = Groups & statistics
single.input = 1
single.dataType = LDF segments (segmentedLDF, segmentedTime)
single.t0 = 0
single.baseline = [-5 0]
single.direction = Auto
single.features = {Peak latency, Onset delay (50%), FWHM, AUC positive, AUC negative, Rise time, Decay time, Peak amplitude, Stim-response integral}
single.series = Trial 1
single.extracted = true
groups.files = [1x24 struct]
groups.order = {Control, Stimulated, Drug}
groups.groupName = Drug
groups.feature = Peak amplitude
groups.subjectMode = File (mean trace)
groups.t0 = 0
groups.baseline = [-5 0]
groups.baselineAuto = true
groups.direction = Auto
groups.design = Paired (same animals)
groups.method = Parametric
groups.groupA = Control
groups.groupB = Stimulated
groups.plotStyle = Mean ± SEM
groups.tested = true

KEY RESULTS
Single file: 8 series (LDF segments)
PeakLatency_s: mean 4.225, SD 0.4367 (n = 8)
OnsetDelay_s: mean 2.025, SD 0.1982 (n = 8)
FWHM_s: mean 5.162, SD 0.7558 (n = 8)
AUCpos: mean 181.6, SD 21.23 (n = 8)
AUCneg: mean 15.96, SD 6.034 (n = 8)
RiseTime_s: mean 2.425, SD 0.44 (n = 8)
DecayTime_s: mean 3.062, SD 0.6718 (n = 8)
PeakAmp: mean 30.85, SD 2.492 (n = 8)
Integral: mean 2616, SD 63.09 (n = 8)
Groups: Control (8 files), Stimulated (8 files), Drug (8 files)
Group test: Paired t-test: t(7) = 10, p < 0.0001; d_z = 3.55;
Stimulated - Control = 11.9 [95% CI 9.12, 14.7]; n = 8 pairs.